Detection of Atmospheric Blocking and Northeast China Cold Vortex Using Satellite Microwave Temperature-Sounding Observations

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***Abstract*—The occurrence of atmospheric blocking signifies a remarkable transition from zonal to meridional circulation at 500 hPa. Its cutoff low may evolve into a northeast China cold vortex (NCCV), causing extreme weather, such as heavy rainfall and hailstorms. In this study, satellite-based microwave temperature-sounding data are used for tracking NCCV movements and for detecting blocking occurrences. The NCCV movement tracks are identified by the warm centers of brightness temperature measurements of the sounding channel, whose peak weighting function (WF) is located at 200 hPa. For the detection of blocking, a linear regression is first established between 500-hPa geopotential height and advanced microwave sounding unit-A (AMSU-A) channels 4–8; and the blocking index is then calculated for the identification of blocking. Here, the block index is calculated as the difference in 500-hPa geopotential height between 40°N and 60°N. Time-longitude plots of the blocking index, together with sequences of 500-hPa-height charts, for two blocking events that occurred in May 2023 and August 2020, compared well with those obtained by the ERA5 global reanalysis. The cutoff low of both blockings developed into NCCVs. This enables operational forecasters to directly and conveniently extract geopotential height information from satellite raw measurements of brightness temperature for weather diagnosis and forecasting. It also provides novel methodologies for analyzing of atmospheric large-scale circulation in recent decades, given more than 26 years of satellite-based microwave temperature-sounding data globally.**

***Index Terms*—Advanced microwave sounding unit-A (AMSU-A), blocking index, northeast China cold vortexes (NCCVs).**

I. Introduction

A POLAR-ORBITING environmental operational satellite (POES) carrying a microwave temperature sounder provides global satellite-based microwave temperature-sounding observations twice daily. Since 1978, USA sequentially launched the Television and Infrared Observation Satellite (TIROS-N), NOAA-6, −7, …, −14 carrying the microwave temperature sounders named MSU of four channels; NOAA-15, …, −19 carrying the microwave temperature sounders named advanced microwave sounding unit-A (AMSU-A) of 15 channels; and the Suomi national polar-orbiting partnership (S-NPP), NOAA-20, and NOAA-21 carrying the microwave temperature sounders named ATMS of 16 temperature channels. Starting in 2006, the European Organization for the exploitation of meteorological satellites (EUMESAT) launched successively MetOp-A, -B, and -C carrying AMSU-A. In 2008, China started the second generation Fengyun (FY) series; the first two satellites FY-3A/B were equipped with MWTS that was similar to MSU [1], [2]; the FY-3C and FY-3D were equipped with MWTS-2 of 13 channels the same as AMSU-A channels 3–15 [3]; and the FY-3E was equipped with MWTS-3, which included 14 channels the same as AMSU-A channels 1–14 (see [4, Table I]). The FY-3E has an early-morning-orbit and was launched on July 5, 2021. FY-3E, MetOp-C, and FY-3D together give a simultaneous three-orbit constellation of early-morning orbit, the mid-morning orbit, and the afternoon orbit, respectively, to provide global coverages of satellite-based microwave temperature-sounding data six times daily.

A northeast China cold vortex (NCCV) is defined as a vortex system that has a closed contour of geopotential height and a cold center at 500 hPa in the region of 35°N–60°N, 105°E–145°E, and maintained for more than three days [5], [6]. It is a large-scale weather system and can induce heavy rainstorms along its path. Niu et al. [7] and Liu and Zou [8] directly applied satellite-based microwave temperature-sounding data to investigations of NCCV. Based on the temperature retrievals from FY-3D MWTS-2 and MetOp-C AMSU-A brightness temperature measurements, Niu et al. [7] reported that NCCVs have warm cores in the upper troposphere and low stratosphere and cold cores in the middle and low troposphere. Liu and Zou [8] showed a feasibility to use FY-3D MWTS-2 upper-level channels for a near-real-time monitoring of NCCV moving tracks. This study develops an objective methodology to determine NCCV moving tracks with brightness temperature measurements from MWTS-3, AMSU-A, and MWTS-2 of the three-orbit constellation of FY-3E, MetOp-C, and FY-3D, respectively.

The NCCV is a type of cutting-off cold low-pressure system. A cutoff low is formed by a deepening of the longwave trough toward the south, which causes an accumulation of cold air in the southern part of the trough. In contrast, a continuous northward extension of longwave ridge in the westerly belt leads to a formation of warm high-pressure system due to an

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accumulation of warm air in the northwestern part of the ridge. A warm high-pressure and a cold cutting-off low pressure often occur in pairs to form the so-called blocking [9], [10]. Blocking is a low-frequency system that can maintain for a long period of time [11]. In the Asian region, blocking appears more commonly in the Ural Mountains and Okhotsk Sea areas and has a duration of 3–8 days [12].

In the northern hemisphere, the annual high-incidence regions of blocking phenomena are mainly concentrated in the East Atlantic through Europe to Central Asia and the central and eastern Pacific regions; the cutoff lows in the northern hemisphere are more commonly found in Southern Europe and the eastern Atlantic coast, China-Siberia, the eastern North Pacific, and western North America; blocking patterns often appear in the eastern Lake Baikal, Ural Mountains, and Sea of Okhotsk regions [12], [13], [14], [15]. The blocking has a profound impact on weather and climate systems [16]. The phenomenon of atmospheric blocking was first discovered by Garriott in 1904 [17]. It is a key driving factor for extreme weather in middle latitudes [18]. It induces circulation anomalies, alters the transport and distribution of heat and water vapor, and triggers extreme weather events, such as large-scale cold waves, high-temperature heat waves, precipitation anomalies, and persistent cold anomalies in East Asia [19], [20], [21], [22], [23]. The cutting-off low, which often forms as part of blocking system, can trigger sustained precipitation and affect the amount and distribution of precipitation [9]. The long-term persistency of such abnormal weather can thus have a profound impact on hemispheric climate, changing the average state and variability of climate [18]. In addition, atmospheric blocking is closely related to the North Atlantic Oscillation and the Pacific North American teleconnection type [22], [23]. Being an important manifestation of anomalous atmospheric circulation [24], the prediction of blocking and cutting-off low is a key to improve extended period weather forecasting and short-term climate prediction [25].

In order to identify blocking phenomena and quantitatively describe their characteristic activities, various blocking indices had been proposed [26], [27], [28], [29], [30]. All blocking indices are calculated from the 500-hPa geopotential height fields of analysis and reanalysis datasets, because the 500-hPa isobaric surface is less affected by the surface and can clearly reflect the blocking occurrence and maintenance [31], [32]. The abovementioned studies employed either reanalysis datasets or numerical simulations of the atmosphere from models of different complexities. In this study, we attempt to examine the atmospheric blocking signals embedded in satellite microwave temperature sounding observations of brightness temperature. A method is proposed for directly calculating the blocking index based on satellite-based microwave temperature-sounding data of brightness temperature, whose feasibility is verified by comparing our results with those obtained from the ERA5 reanalysis. POES microwave temperature sounders measure the upwelling microwave radiation emitted primarily from atmospheric oxygen. Since oxygen is nearly uniformly and stably distributed through atmosphere, their sounding channels located at and above the middle troposphere are mainly functions of atmospheric temperature. The microwave temperature-sounding channel observations were successfully applied for detecting global warming trends in the troposphere and cooling trends in low stratosphere [33], [34], [35], as well as well-known teleconnection patterns: 1) eastern Atlantic pattern; 2) Pacific/North American pattern; 3) western Atlantic pattern; 4) western Pacific pattern; and 5) Eurasian pattern. Xia et al. [36] reported a relationship between AMSU-A observed brightness temperature at window channels and sea surface temperature in the tropical oceans. A brightness-temperature-based Niño Index successively detected six of the seven El Niño cases that occurred between 1998 and 2020 except for the very weak one in 2003. These results call for further exploration of the direct application of satellite observations in weather and climate research.

Satellite observations were used to retrieval many atmospheric variables. For examples, microwave temperature sounding observations were used to retrieve hurricane warm core structures using traditional regression methods [37], [38] and advanced machine learning neural network models [39]. Multiple cloud variables, including cloud mask, phase/type, cloud-top height, optical thickness, and particle effective radius, were retrieved from the visible and infrared measurements of the Advanced Himawari Imager (AHI) on board the Japanese geostationary satellites Himawari-8 and Himawari-9 [40]. Wang et al. [41] retrieve cloud optical thickness and cloud-top height by combining data from the MODIS imager aboard the Aqua satellite and the CALIPSO satellite using a deep learning model, achieving superior performance in day-night retrieval consistency and computational efficiency. Wang et al. [42] further expanded the deep learning model's functionality to retrieve cloud base height and cloud thermodynamic phase states, improving multilayer cloud identification and phase recognition. In this study, AMSU-A microwave brightness temperature observations from MetOp-C are used to retrieve geopotential height at 500 hPa. Directly retrieving 500-hPa geopotential height from satellite observations has never been done before. In a follow-on study, we plan to apply machine learning methods for retrieving geopotential height fields.

II. Data Description

The microwave temperature sounders used in this study include AMSU-A onboard MetOp-C, MWTS-2 onboard FY3D, and MWTS-3 onboard FY-3E. The equator crossing time of FY-3E, MetOp-C, and FY-3D is 5:30 LST, 9:30 LST, and 2:00 LST, respectively. Table I presents a comparative analysis of their channel numbers, central frequencies, and peak weighting function (WF) levels. AMSU-A is a microwave radiometer carried by NOAA-15,..., –19 and MetOp-A, -B, and -C and employing a cross-track scanning method. Each scan line of AMSU-A has 30 fields of views (FOVs). FOVs 15 and 16 are near nadir and FOVs 1 and 30 are located at two swath edges with scan angles of $\pm 49.5°$. The diameter of FOV is the smallest at the nadir (~48 km) and the largest at the swath edges (~155 km). AMSU-A has 12 temperature sounding channels (channels 3–14) and three window channels (channels 1, 2, and 15). Channels 3–14 are located within the 50–58-GHz oxygen band and

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TABLE I

CH NUMBERS, CENTER FREQUENCIES, AND PRESSURE LEVELS OF PEAK WFS OF AMSU-A, MWTS-2, AND MWTS-3. THE BOXED AND UNDERLINED FREQUENCIES ARE THE FREQUENCIES OF THE CHANNELS SPECIFIC TO MWTS-2 AND MWTS-3, RESPECTIVELY

Channel Number			Frequency (GHz)	Maximum WF Level (hPa)
AMSU-A	MWTS-2	MWTS-3		
1			23.8	1085
2			31.4	1085
3	1		50.3	1085
	2	4	51.76	900
4	3	5	52.8	850
		6	53.246	700
5	4	7	53.596	650
		8	53.948	565
6	5	9	54.40	400
7	6	10	54.94	250
8	7	11	55.50	200
9	8	12	57.290344(f_0)	100
10	9	13	$f_0 \pm 0.217$	50
11	10	14	$f_0 \pm 0.3222 \pm 0.048$	25
12	11	15	$f_0 \pm 0.3222 \pm 0.022$	10
13	12	16	$f_0 \pm 0.3222 \pm 0.010$	5
14	13	17	$f_0 \pm 0.3222 \pm 0.0045$	2
15			89	1085

can provide temperature sounding data from the near surface layers to the low stratosphere, while channels 1, 2, and 15 are used to obtain information on clouds and water vapor in the atmosphere as well as and the Earth's surface parameters [43]. This study focuses on microwave temperature sounding channels.

MWTS-2 maintains channel configuration consistency with AMSU-A sounding channels except for adding a low tropospheric channel around 900 hPa. Specifically, it has 13 sounding channels, in which channels 1 and 3–13 are similar to AMSU-A channels 3–14 and channel 2 was newly added to enhance sounding capability in the lower troposphere. It is worth mentioning that MWTS-2 does not have any window channels.

MWTS-3 is a successor of MWTS-2. The frequencies of MWTS-3 channels 1–3, 5, 7, and 9–17 are identical to AMSU-A channels 1–14. MWTS-3 has three additional sounding channels 4, 6, and 8, whose central frequencies are 51.76, 53.246, and 53.948 GHz, respectively; and peak WFs are located at 900, 700, and 565 hPa, respectively. The three new sounding channels significantly enhance the observational capabilities of the middle and lower troposphere, which is crucial for numerical weather prediction (NWP).

The microwave observations of brightness temperature from three-orbits constellation of FY-3E, MetOp-C and FY-3D are used for directly tracking the NCCV trajectories. The specific channel used for doing so is the sounding channel, whose center frequency is 55.5 GHz and peak WF is located at 200 hPa, which correspond to channel 8 of AMSU-A, channel 7 of MWTS-2, and channel 11 of MWTS-3. Together, the three microwave temperature sounders, FY-3E MWTS-3, MetOp-C AMSU-A, and FY-3D MWTS2, provide global observations six times daily at intervals varying from 3.5 to 4.5 h.

The fifth-generation reanalysis dataset was constructed by the European Centre for Medium-Range Weather Forecasts (ERA5). It assimilated multisource observations based on advanced NWP models. This dataset provides hourly estimates of physical quantities related to the atmosphere, ocean, and land surface [44]. In this study, hourly geopotential height data on the 500-hPa isobaric surface at a horizontal resolution of $0.25° \times 0.25°$ were used for determining atmospheric blocking as well as for retrieving the 500-hPa geopotential height directly from brightness temperature measurements of AMSU-A channels 4–8 through linear regression method.

III. METHODOLOGY DESCRIPTION

A. NCCV Center Positioning Method

The AMSU-A, MWTS-2, and MWTS-3 are cross-tracking radiometers. At the nadir, the optical path detected by these radiometers is the straight line connecting the instrument and the nadir and is of the shortest path length. As the FOV gradually moves away from the nadir at the same scan line, the scan angle increases, and thus, the optical path length continuously increases. In the troposphere where the atmospheric temperature decreases with increasing altitude, the larger the scan angle, the lower the brightness temperature. In contrast, in the stratosphere where the atmospheric temperature increases with increasing altitude, a larger scan angle leads to a higher brightness temperature [45]. This phenomenon is called the limb effect, which significantly interferes with an effective identification of weather signals in the brightness temperature distributions. In order to directly use observed brightness temperature data for the NCCV center positioning, it is necessary to perform limb correction on brightness temperature measurements to improve their usability.

This study adopts the limb correction method that was proposed by Goldberg et al. [46] and carefully described by Zou and Tian [38]. The main idea is to utilize the observed brightness temperature of adjacent channels to perform limb correction on the observed brightness temperature of the target channel. The limb corrected brightness temperature is calculated as follows:

$$T_{b,k}^{LC}(\lambda,\varphi;\alpha_i,\varphi_{\text{nadir}}) = b_i + \sum_{k_p} a_{k_p,i}\left(\overline{T_{b,k_p}^{\text{obs}}(\lambda,\varphi;\alpha_i,\varphi_{\text{nadir}})} - \overline{T_{b,k_p}^{\text{obs}}(\varphi_{\text{nadir}})}\right) \quad (1)$$

where the subscript "k_p" represents the adjacent channels used for the limb correct of the target channel k, (λ,φ) is the location of $T_{b,i}^{\text{obs}}$, α_i is the scan angle of $T_{b,i}^{\text{obs}}$, φ_{nadir} is the nadir latitude of the same scan line as $T_{b,i}^{\text{obs}}$, $\overline{T_{b,k_p}^{\text{obs}}(\varphi)}$ represents the global mean of the observed brightness temperatures near the nadir (i.e., AMSU-A FOVs 14 and 15), $\overline{T_{b,k_p}^{\text{obs}}(i,\varphi)}$ represents the global mean of the observed brightness temperatures at the ith FOV (i.e., AMSU-A FOVs 1–13 and 16–30), and b_i and $a_{k_p,i}$ are the limb correction coefficients. $k_p$in (1) takes the values of 7–9 for the AMSU-A target channel 8, 6–9 for the MWTS-2 target channel 7, and 10–13 for the MWTS-3 target channel 11. The maximum brightness temperature value after limb correction, $T_{b,k}^{LC}(\text{FOV}_i,\varphi)$, is taken as the NCCV center

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position. The values of regression coefficients are obtained by a least square fit to the global mean brightness temperatures observed near the nadir using one-month training data prior to the studying case of NCCV.

When there are overlapping swaths in high latitudes, the latest time AMSU-A swath data is used, so that the NCCV centers from a single POES are determined twice daily. In this study, three NCCV cases that occurred in 2020 and 2023 were investigated. The NCCV area is defined as 35°N–60°N and 115°E–145°E. The NCCV center positioning method based on satellite brightness temperature data consists of the following three steps.

1) Screen out the brightness temperature data within the longitude range of 60°E–145°E and the latitude range of 30°N–70°N and interpolate these data onto the regular grid points at 0.25° × 0.25° horizontal resolution. Compare the brightness temperature value of each grid point with the brightness temperature values of the surrounding grid points. When the brightness temperature value of the central grid point is higher than those of the surrounding grid points, the central grid point is a maximum value point.
2) If there is only one extreme value point at a certain time, then consider this point as the NCCV center. If there are multiple maximum points at a certain time, select the maximum point closest to the NCCV center determined at the previous time as the vortex center at the current moment.
3) If there are multiple maximum points at the initial time, use the method in Step 2) to trace the NCCV centers backward. Continue to select the maximum point at the previous time that is closest to the next-time NCCV center until the initial time.

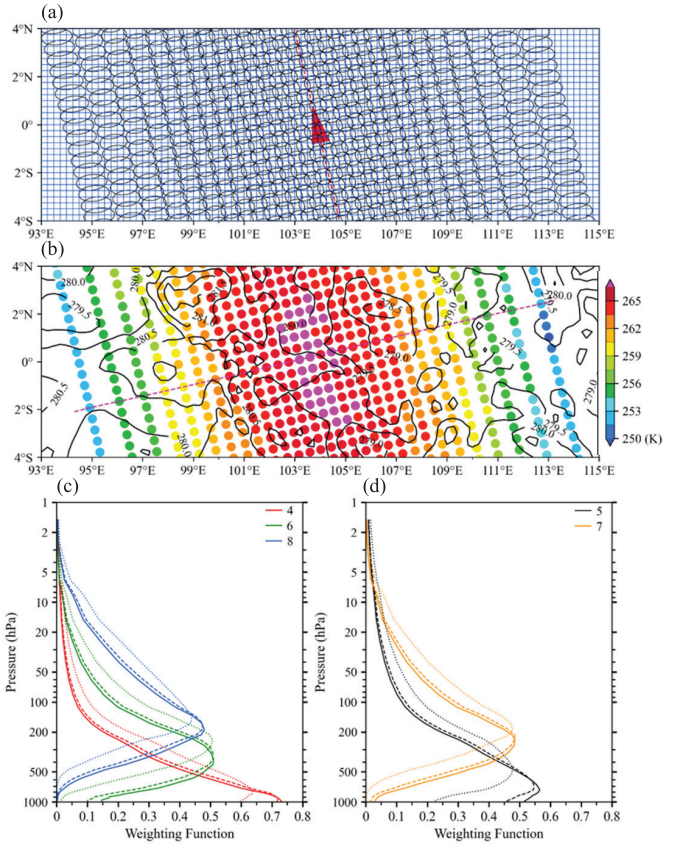


Fig. 1. Spatial distributions of (a) AMSU-A FOV (black circle) at MetOp-C ascending node (orbital direction in red arrow) on May 9, 2023 and ERA5 grids (blue line). (b) Spatial distribution of all-sky simulation of channel-5 brightness temperatures (shaded in color) and ERA5 500-hPa temperature (contour interval: 0.5 K). WFs of (c) channels 4, 6, and 8 and (d) channels 5 and 7 FOVs (dotted line) 1, (solid line) 8, and (dashed line) 15 along the scan line [magenta dashed curve in (b)].

B. Regression Model for 500-hPa Geopotential Height

Some basic information of AMSU-A channels 4–8 from MetOp-C are displayed in Fig. 1. Specifically, Fig. 1(a) presents the spatial distributions of FOV sizes (black circle) of AMSU-A brightness temperature measurements [see Fig. 1(b)] near the equator on May 9, 2023, along with the ERA5 grid lines (blue line). As the scan angle of FOV increases, the ratio of the major axis to the minor axis of the elliptical FOV increases. This leads to denser and higher resolution AMSU-A observations near the nadir. In order to show the high dependence of brightness temperature on the scan angle due to the limb effect, we show in Fig. 1(b) the spatial distributions of all-sky simulations of AMSU-A channel-5 brightness temperature (shaded in color) and 650-hPa temperature (contour) of the ERA5 dataset. Here, all-sky simulations are generated using the ERA5 reanalysis as an input to the fast radiative transfer model RTTOV13.0 [47]. There is a difference between brightness temperature and temperature distribution pattern. The simulated brightness temperatures near the nadir are significantly higher than those at the larger scan angles of the swath. In addition, due to the latitudinal differences in solar radiation, the brightness temperature varies with latitude. Therefore, we set the brightness temperature and geopotential height as functions of scan angle and latitude. Fig. 1(c) and (d) shows the WFs of channels 4–8 obtained using the RTTOV13.0. The peak heights of WFs of channels 4–8 varying with scan angle. The WFs of these five sounding channels measure the atmospheric radiation in the troposphere.

Being located within the oxygen absorption band ~50–58 GHz, which are much smaller than 200 GHz, microwave temperature sounding channels brightness temperatures are approximately linear functions of the atmospheric temperature [7]. Therefore, the atmospheric temperature at a specific pressure level can thus be retrieved by performing a linear regression expressing it as a weighted sum of the brightness temperature measurements of different sounding channels, whose WF peaks are near this pressure layer [37]. In hydrostatic condition, differences of geopotential height (Z) between two pressure layers (p_1 and p_2) are proportional to the layer-mean temperature

$$Z_2 - Z_1 = \frac{R_d}{g}\int_{p_2}^{p_1}\frac{T}{p}dp \approx \langle T\rangle\frac{R_d}{g}\ln\left(\frac{p_1}{p_2}\right) \tag{2}$$

where R_d is the dry-air gas constant. Therefore, we can similarly express the geopotential height on the 500-hPa

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isobaric surface as a weighted sum of the observed brightness temperatures of AMSU-A channels 4–8

$$Z_{500}^{\text{sat}}(\lambda,\varphi) = C_0(\alpha_i,\varphi_{\text{nadir}}) + \sum_{i=4}^{8} C_i(\alpha_i,\varphi_{\text{nadir}})\, T_{b,i}^{\text{obs}}(\lambda,\varphi;\alpha_i,\varphi_{\text{nadir}}) \quad (3)$$

where the subscript "i" represents the channel number, $T_{b,i}^{\text{obs}}$ represents the brightness temperature observation at the ith channel, (λ,φ) is the location of $T_{b,i}^{\text{obs}}$, α_i is the scan angle of $T_{b,i}^{\text{obs}}$, φ_{nadir} is the nadir latitude of the same scan line as $T_{b,i}^{\text{obs}}$, and C_i (i = 0, 4, 5,..., 8) are the regression coefficients. The values of regression coefficients are obtained by a least square fit to the ERA5 500-hPa geopotential height during the two-week training period from April 25, 2023, to May 8, 2023.

C. Blocking Index Calculation

The occurrence of blocking is declared if both $I(\lambda)$ and $\bar{I}(\lambda)$ are less than zero, where $I(\lambda)$ and $\bar{I}(\lambda)$ are defined as [26]

$$I(\lambda) = Z_{500}(\lambda, 40°\text{N}) - Z_{500}(\lambda, 60°\text{N}) \quad (4)$$

$$\bar{I}(\lambda) = \frac{1}{3} I(\lambda - 10°) + I(\lambda) + I(\lambda + 10°). \quad (5)$$

In other words, the blocking index is defined by the difference of geopotential height at 500 hPa between the lower latitude (φ_l = 40°N) and the higher latitude (φ_h = 60°N), $I(\lambda)$; as well as the average of $I(\lambda)$ within the 20° longitudinal range centered at λ, $\bar{I}(\lambda)$. When blocking occurs, the indices $I(\lambda)$ and $\bar{I}(\lambda)$ are less than zero, indicating that the geopotential height at 500 hPa at the lower latitude is lower than the higher latitude.

IV. Results

A. Determining NCCV Moving Paths Directly From Three-Orbit Microwave Observations

The moving paths of the NCCV case that lasted from July 11 to 16, 2023 are determined based on brightness temperature measurements from MWTS-3, AMSU-A, and MWTS-2 onboard the three-orbit constellation of FY-3E, MetOp-C, and FY-3D, respectively. Due to the downward intrusion of the stratospheric air, there are two outstanding characteristics at NCCV locations: 1) the tropopause is lower than the sounding environment, below 200 hPa and 2) there exists a warm core centered at 200 hPa (see Fig. 12 in Niu and Zou [7]). Fig. 2(a), (c), and (e) shows the spatial distributions of brightness temperature measurements of MWTS-3 channel 11, whose peak WF is located near 200 hPa, at 2340 UTC on July 11, 2321 UTC on July 12, and 1011 UTC in July 2023, respectively. In Fig. 2, areas where the brightness temperature is less than 225.5 K are shaded in color, and areas where the brightness temperature is greater than 225.5 K are indicated by black contours at 1-K interval. The NCCV center identified from MWTS-3 observations is marked with a white cross symbol. The NCCV area is indicated by black dashed lines of a trapezoid shape. The observation time at the center of the plotted domain (50°N, 90°E) is given above each panel on the left. It can be seen that the center of NCCV does locate at

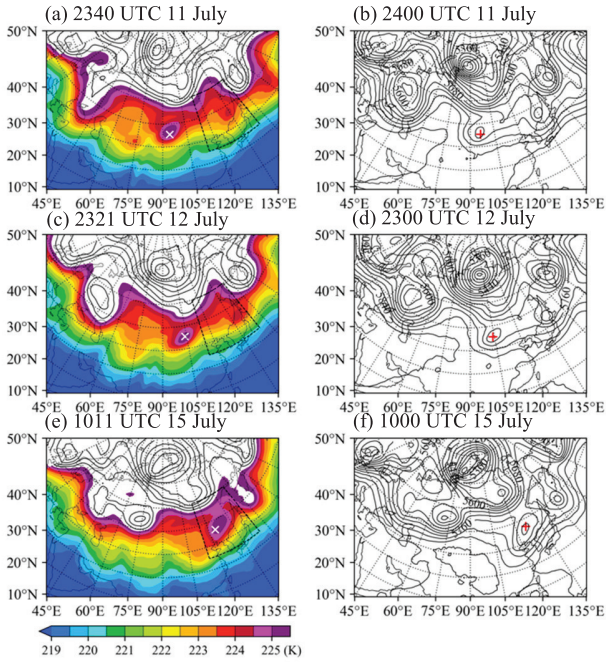


Fig. 2. Spatial distributions of FY-3E MWTS-3 channel-11 (55.50 GHz) limb-corrected brightness temperature observations (left panels) and the ERA5 500-hPa geopotential height analysis (right panels, black contours at 40-gpm intervals) on (a) and (b) July 11, (c) and (d) July 12, and (e) and (f) July 15, 2023. The observing times of brightness temperature at the center domain point (50°N, 90°E) and the analysis times are indicated above each panel. Brightness temperatures smaller and greater than 225.5 K are shown in color shading and black curves (contour interval: 1 K), respectively. Also shown are the NCCV center locations determined by brightness temperature observations (white cross symbol) and 500-hPa geopotential height (red plus sign), as well as the NCCV domain (black dashed box).

the maximum brightness temperature of MWTS-3 channel 11. At 2340 UTC on July 11, a warm core was separated from the brightness temperature trough, forming an isolated system [see Fig. 2(a)]. The next day, it moved eastward and approached the NCCV area by 2321 UTC on July 12 [see Fig. 2(c)]. After 1049 UTC on July 13, the warm core was located within the NCCV area and reached its maximum intensity [see Fig. 2(e)]. The NCCV weakened afterward (Figures omitted). The large-scale distribution patterns of brightness temperatures (left panels in Fig. 2) resemble very much to those of the ERA5 500-hPa geopotential height fields (left panels in Fig. 2) in the middle and high latitudes. Regions with higher brightness temperatures correspond to the regions of lower geopotential heights at 500 hPa. Specifically, the isolated warm core seen in brightness temperature measurements [see Fig. 2(a), (c), and (e)] coincides with the geographical location of the cutoff low on the 500-hPa geopotential height fields [see Fig. 2(b), (d), and (d)]. The red plus sign in Fig. 2(b), (d), and (d) represents the center of the cutoff low.

A single POES carrying a microwave temperature sounder can only provide twice-daily global coverage of brightness temperature measurements. By combining the early-morning orbiting FY-3E with the morning-orbiting MetOp-C and the afternoon-orbiting FY-3D, an NCCV can be observed six times daily. Detection results of the same NCCV shown in Fig. 2 by the three-orbit POES constellation are presented in

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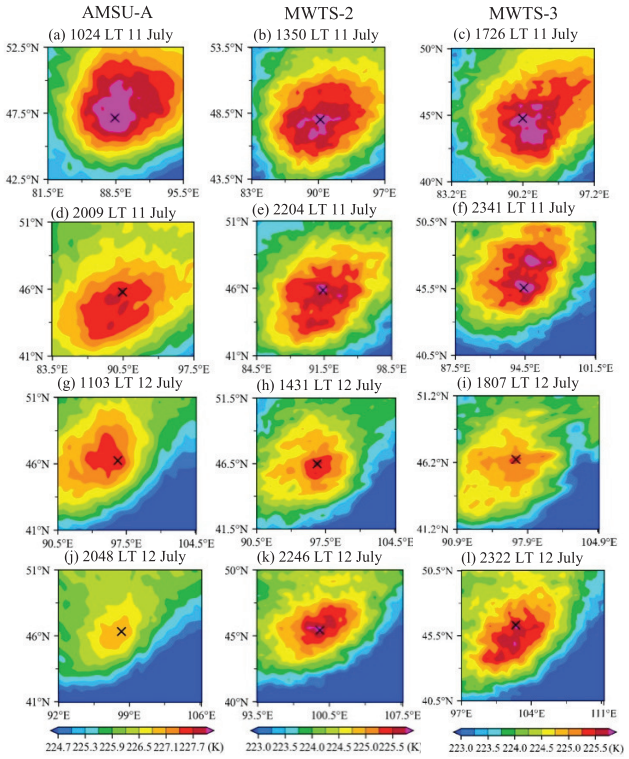


Fig. 3. Spatial distributions of MetOp-C AMSU-A channel 8 (first column), FY-3D MWTS-2 channel 7 (second column), and FY-3E MWTS-3 channel 11 (third column) brightness temperature observations at six times on (a)–(f) July 11 and (g)–(l) July 12, 2023. The NCCV centers and LTs determined by TB observations are indicated by black crosses and above each figure panel, respectively. The center frequency of all three channels is 55.50 GHz.

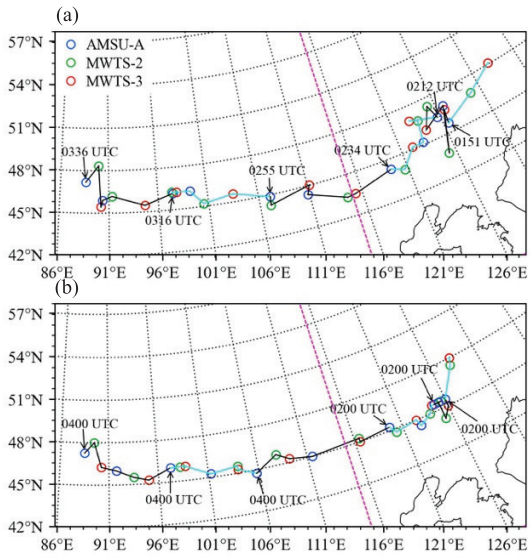


Fig. 4. (a) NCCV track that occurred in July 11–16, 2023 determined by brightness temperature observations of (blue) MetOp-C AMSU-A, (green) FY-3D MWTS-2, and (red) FY-3 E MWTS-3, in which the observation position and time at the beginning of each day are also marked on the figure and the lines connecting NCCV centers are indicated alternatively in black and cyan every other day. (b) Same as (a) except based on the 500-hPa geopotential height of the ERA5 reanalysis. The west edge of the NCCV area is indicated by dashed line in magenta color.

Figs. 3 and 4. To present the diurnal change of the NCCV intensity more clearly, Fig. 3 shows the warm core distributions near the NCCV centers in a small area of 10° latitudes and 6° longitudes, which are observed by the 55.5-GHz channels of AMSU-A, MWTS-2, and MWTS-3 on July 11 and 12, 2023. The center of the NCCV determined by the maximum value of the brightness temperature is represented by a black cross, and the local time (LT) when the satellite scans over the center of the NCCV is marked above each subfigure. During the two-day period, the NCCV moved approximately 15 longitudes eastward. The intensity of warm core exhibits diurnal variation, with the strongest and weakest times on July 11 being 1024 and 2009 LT, respectively; and the strongest and weakest times on July 12 being 1103 and 2048 LT, respectively. The NCCV experienced an overall weakening from July 11 to 12 before entering the NCCV area.

Fig. 4 shows the movement path of the NCCV from July 11 to July 16, 2023. The path in Fig. 4(a) is determined by the brightness temperature measurements of three-orbit Fof FY-3E, MetOp-C, and FY-3D. The blue, green, and red circles represent the positions of the NCCV centers determined by the observed brightness temperatures of MetOp-C AMSU-A, FY-3D MWTS-2, and FY-3E MWTS-3, respectively. The path in Fig. 4(b) is determined according to the hourly geopotential height fields at 500 hPa of the ERA5 reanalysis. When screening the reanalysis data, we ensure that the analysis time is closest to the observation time in Fig. 4(a). In Fig. 4, the observation positions and times at the starting time of each day are marked, and the lines connecting the centers of the NCCV are alternately represented by black and cyan every other day. By comparing Fig. 4(a) and (b), we find that the movement paths of the NCCV determined by two different data types are roughly the same. The NCCV moves northeastward as a whole, and both enter the NCCV area (115°E–140°E, 35°N–60°N) on July 13, 2023. On July 15, the NCCV path was limited to adjacent local area. After July 16, closed contour lines can no longer found in the 500-hPa geopotential height fields nor the horizontal distributions of observed brightness temperatures, near the NCCV positions in the previous times. Therefore, brightness temperature measurements of the channel of microwave temperature sounders whose peak WF at 200 hPa from a three-orbit POES constellation can successively track an NCCV trajectory at intervals of about 4 h.

B. 500-hPa Geopotential Height Retrieval From AMSU-A Channels 4–8 Observations

AMSU-A temperature sounding channels 4–8 measure thermal radiation emitted from different layers of the atmosphere from around 700 to 200 hPa. Taking the MetOp-C AMSU-A observations of brightness temperature from April 25 to May 8, 2023 as the training dataset, the regression model expressing the 500-hPa geopotential height as a weighted sum of brightness temperature measurements from channels 4 to 8 is first established following the method described in Section III-B. We then utilize the brightness temperature measurements from May 9 to 23, 2023 to obtain the geopotential height field during the same period using the regression model.

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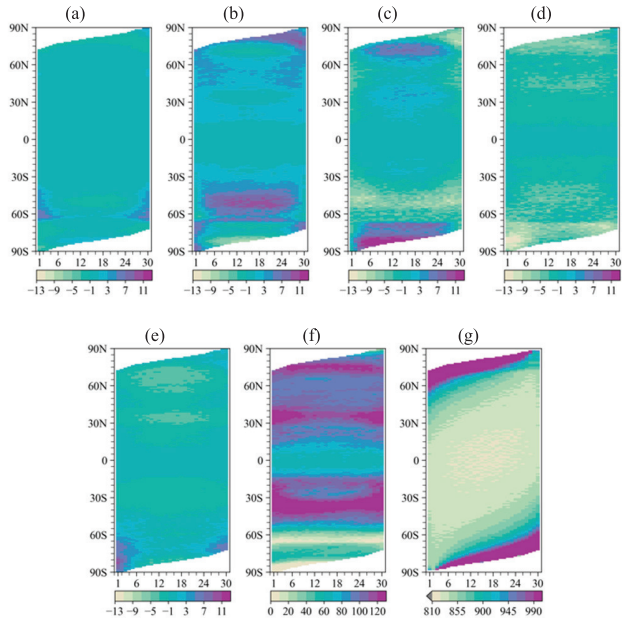


Fig. 5. (a)–(f) Latitudinal variations of the regression coefficients for channels 4–8 and the intercept with respect to FOV numbers (x-axis). (g) Data count within 1° latitudinal band for retrieving Z_{500}^{sat} from MetOp-C AMSU-A brightness temperature observations during the training period from April 25 to May 8, 2023.

First, we show the variations of the five regression coefficients for channels 4–8 [see Fig. 5(a)–(e)] and the intercept [see Fig. 5(f)] with respect to FOV number and latitude. The distribution of regression coefficient of each predictor channel reflects a latitudinal dependence of relative contributions of brightness temperatures at different FOVs to the 500-hPa geopotential height. All regression coefficients and the intercept exhibit a characteristic scan-angle symmetry with respect to the nadir. Variations with respect to latitude have banded structures in terms of the tropics, subtropics, middle latitudes, and high latitudes. The data counts of training sample at the 1° latitudinal interval are provided in Fig. 5(f). The reason for the larger amount of data in high latitudinal regions is that the inclination of MetOp-C (98.7°) is greater than 90°, resulting a large number of overlapping swaths there.

Second, we present the spatial distributions of retrieved 500-hPa geopotential height (Z_{500}^{sat}) and AMSU-A brightness temperature of channel 5 ($T_{b,\mathrm{channel5}}^{\mathrm{obs}}$) around 1545 UTC on May 12, 0315 UTC on May 15, 1520 UTC on May 18, and 0431 UTC on May 21, 2023 (Fig. 6). The center positions of NCCV on May 18 and 21 are marked with black crosses. It can be seen that the spatial distribution of $T_{b,\mathrm{channel5}}^{\mathrm{obs}}$ is in remarkable agreement with the circulation pattern of Z_{500}^{sat}. The zonal circulation near the western side of the Ural Mountains transformed into a meridional circulation. There is a low-pressure trough downstream of the high-pressure ridge, and the trough in the westerlies continuously deepens and develops, causing high latitude cold air continuously accumulating into the trough. The brightness temperature trough coexists with the geopotential trough. The southern part of the upstream high-pressure ridge is gradually invaded by cold air, which weakens

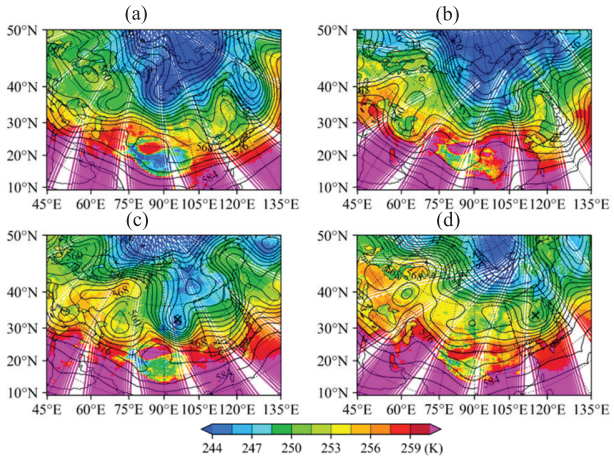


Fig. 6. Spatial distributions of $Z_{500}^{\mathrm{sat}}(\lambda,\varphi)$ (contour interval: 40 gpm) and MetOp-C AMSU-A channel-5 brightness temperature observations (color shading) at (a) 1545 UTC May 12, (b) 0315 UTC May 15, (c) 1520 UTC May 18, and (d) 0431 UTC May 21, 2023. The NCCV center and the NCCV domain are indicated by black cross mark and black dashed box, respectively.

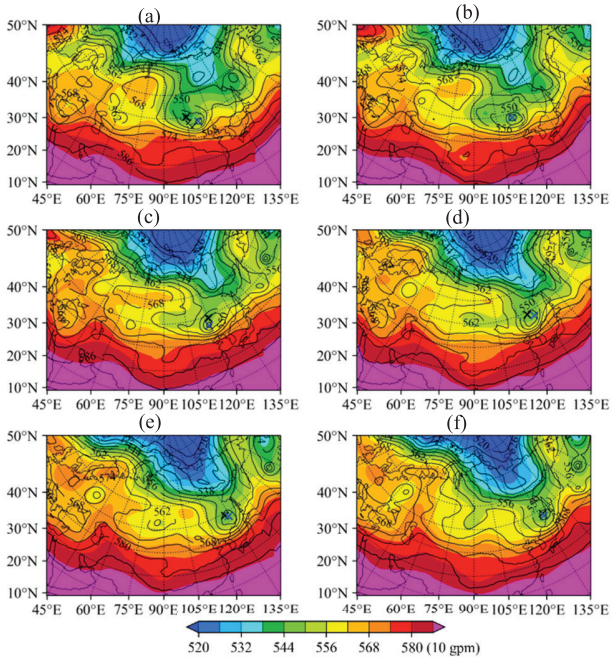


Fig. 7. Spatial distributions of Z_{500}^{sat} (color shading) and Z_{500}^{ERA5} (contours, interval: 60 gpm) during the period from May 19 to 21, 2023. The NCCV centers determined by Z_{500}^{sat} and Z_{500}^{ERA5} are indicated with black and blue crosses, respectively. The times when the MetOp-C AMSU-A passing the location (50°N, 90°E) is indicated above each figure panel, and the reanalysis times of Z_{500}^{ERA5} closest to these times are shown. (a) 0404 UTC May 19. (b) 1457 UTC May 19. (c) 0422 UTC May 20. (d) 1439 UTC May 20. (e) 0431 UTC May 21. (f) 1432 UTC May 21.

the warm air from the south. On May 18, the blocking pattern was seen, along with and the NCCV.

Furthermore, we compare Z_{500}^{sat} with the geopotential height fields from the reanalysis of the ERA5 (Z_{500}^{ERA5}) and the brightness temperature fitting from May 19 to 21, 2023 (Fig. 7). This is a time period during which the blocking high-pressure system and the NCCV appeared simultaneously. The black contour curves display the spatial distribution of Z_{500}^{ERA5}, and

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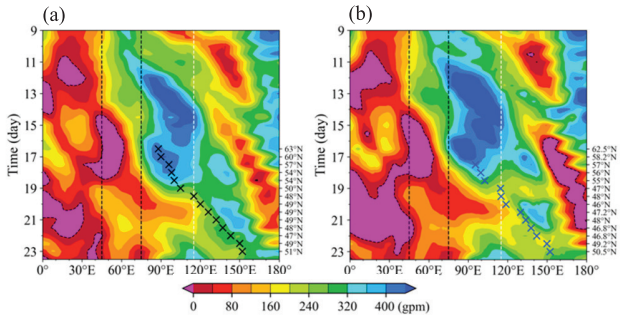


Fig. 8. Temporal variations of (a) (color shading) $I^{\mathrm{sat}}(\lambda)$ and (black cross) NCCV center and (b) (color shading) $I^{\mathrm{ERA5}}(\lambda)$ and (blue cross) NCCV center during the period from May 9 to 23, 2023 at MetOp-C AMSU-A observing times. The longitudes of the Ural and the west edge of the NCCV area are represented by black and white dashed lines, respectively. The black dashed curves in (a) and (b) indicate $I^{\mathrm{ERA5}}(\lambda)$ and $I^{\mathrm{sat}}(\lambda)$ being equal to zero.

color shadings show the spatial distribution of Z_{500}^{sat}. The intervals of the contour curves and color shadings are the same. The NCCV center positions determined by Z_{500}^{sat} are represented by blue crosses, and those determined by Z_{500}^{ERA5} are represented by black crosses. The configurations of trough, ridge, and closed systems of Z_{500}^{sat} and Z_{500}^{ERA5} are comparable. Both fields can clearly reveal the temporal evolution of the blocking high-pressure system and the NCCV. The positions of the NCCV determined by Z_{500}^{sat} and Z_{500}^{ERA5} are very close. As can be seen from Fig. 7, the NCCV moved eastward as a whole and moved into the ocean on May 21 near 50°N and 135°E. During May 19 and 20, the meridional circulation on the western side of the Ural Mountains intensified, its range expands, and an isolate closed circulation formed simultaneously [see Fig. 12(a)–(d)]. On May 21, the range of the blocking high-pressure system and the intensity of the meridional circulation remained stable.

C. Detecting Blocking Events Based on AMSU-A Retrieved Geopotential Height

The blocking index $I^{\mathrm{sat}}(\lambda)$ is calculated at various lon gitudes in the Eastern Hemisphere based on the MetOp-C AMSU-A retrieved geopotential height at 500 hPa from May 9 to 23, 2023 [see Fig. 8(a)]. As a comparison, we also show the temporal and longitudinal variations of $I^{\mathrm{ERA5}}(\lambda)$ [see Fig. 8(b)]. The black dashed line represents the position where the index is equal to 0. The blue and black crosses represent the centers of the NCCV determined based on Z_{500}^{sat} and Z_{500}^{ERA5}, respectively. The longitudinal range of the Ural Mountains is marked by black dashed lines. By comparing $I^{\mathrm{sat}}(\lambda)$ and $I^{\mathrm{ERA5}}(\lambda)$, it is found that there is a significant similarity in their distributions. Both display similar longitudinal range, intensity, and eastward movement characteristics of the blocking high pressure. The absolute extreme values of $I^{\mathrm{sat}}(\lambda)$ and its range are slightly smaller than those of $I^{\mathrm{ERA5}}(\lambda)$. For example, in the region near the Ural Mountains, where $I^{\mathrm{sat}}(\lambda)$ and $I^{\mathrm{ERA5}}(\lambda)$ are small and the blocking high-pressure system occurs and develops, $I^{\mathrm{sat}}(\lambda)$ is slightly larger than $I^{\mathrm{ERA5}}(\lambda)$. Therefore, the blocking threshold based on index $I^{\mathrm{sat}}(\lambda)$ is set to 40 gpm, slightly higher than the threshold of index $I^{\mathrm{ERA5}}(\lambda)$, which is 0. The NCCV appeared after the formation of the blocking high-pressure system. The life cycle of the NCCV was from May 16 to 23, 2023. The longitudes of the NCCV center positions in Z_{500}^{ERA5} and Z_{500}^{sat} fields are relatively consistent, both reflecting the eastward movement path of the NCCV. The cutoff low-pressure system originates from the area around Lake Baikal and then gradually moves eastward and dissipates after reaching 150°E.

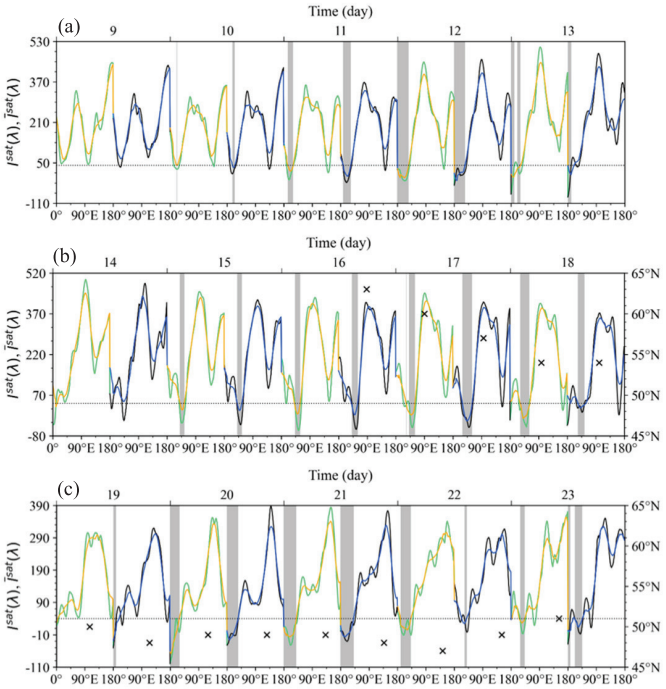


Fig. 9. Variations of (orange and blue) $\bar{I}^{\mathrm{sat}}(\lambda)$ and (green and black) $I^{\mathrm{sat}}(\lambda)$, block occurring longitudinal range (vertical gray shading), and the NCCV center position (blue cross) during the period from (a) May 9 to 13, (b) May 14 to 18, and (c) May 19 to 23, 2023. The descending (ascending) orbits of Metop-C are represented by (blue) orange and (black) green curves. The black dotted line indicates $I^{\mathrm{sat}}(\lambda) = \bar{I}^{\mathrm{sat}}(\lambda) = 40$ gpm.

Fig. 9 presents the daily variations of the two blocking indices of $I^{\mathrm{sat}}(\lambda)$ and $\bar{I}^{\mathrm{sat}}(\lambda)$ within the longitude range of the Eastern Hemisphere from May 9 to 23, 2023, during which there was blocking activity. The black horizontal dotted line represents the threshold line of 40 gpm. When both $I^{\mathrm{sat}}(\lambda)$ and $\bar{I}^{\mathrm{sat}}(\lambda)$ are less than 40 gpm, blocking occurred in the longitudes shaded by gray columns. The center positions of the NCCV are marked with black crosses. Note that the latitudes of NCCV center locations are indicated by the y-axis on the right of Fig. 9. The black and blue curves represent $I^{\mathrm{sat}}(\lambda)$ and $\bar{I}^{\mathrm{sat}}(\lambda)$ at the ascending orbits of MetOp-C, respectively. The green and orange curves show $I^{\mathrm{sat}}(\lambda)$ and $\bar{I}^{\mathrm{sat}}(\lambda)$ at the descending orbits of MetOp-C, respectively. As expected, the peak and valley values of $\bar{I}^{\mathrm{sat}}(\lambda)$ are smaller than those of $I^{\mathrm{sat}}(\lambda)$ due to the smoothing effect within $\pm 10°$ longitudes [see (5)]. Given that blocking high-pressure systems are of large scales, $\bar{I}^{\mathrm{sat}}(\lambda)$ can exclude any possible impacts of small-scale disturbances to blocking detection. This ensures that the detected blocking activities are indeed the large-scale blocking characteristics. For example, there were several times that $I^{\mathrm{sat}}(\lambda)$ reached below 40 gpm at certain longitudes, but $\bar{I}^{\mathrm{sat}}(\lambda)$ remained above 40 gpm, so no large-scale blocking

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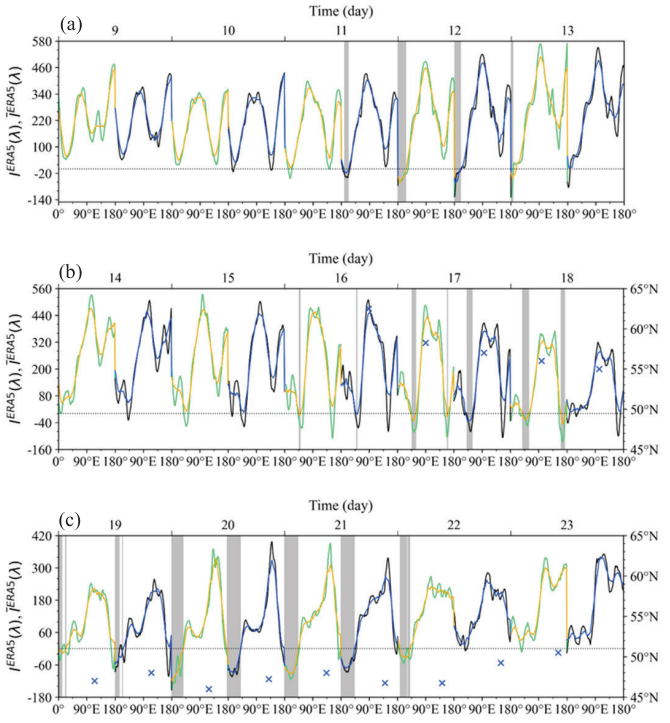


Fig. 10. Variations of (orange and blue) $\bar{I}^{\mathrm{ERA5}}(\lambda)$ and (green and black) $I^{\mathrm{ERA5}}(\lambda)$, block occurring longitudinal range (vertical gray shading), and the NCCV center position (blue cross) during the period from (a) May 9 to 23, (b) May 14 to 18, and (c) May 19 to 23. The descending (ascending) orbits of Metop-C are represented by (blue) orange and (black) green curves. The black dotted line indicates zero line of $I^{\mathrm{ERA5}}(\lambda)$ and $\bar{I}^{\mathrm{ERA5}}(\lambda)$.

was detected. Weak and strong blocking activities are detected before and after May 14, 2023, respectively. The blocking high-pressure system after May 14 was quite stable, lasting for ten days until May 23, 2023. It also triggered the formation of NCCV on May 16, 2023. The activity range of this NCCV is approximately within 45°N–60°N and 90°E–150°E. The NCCV moved southeastward first and turned to the northeast direction after May 21, 2023.

For comparison purpose, we also show the daily variations of the two blocking indices of variations of $I^{\mathrm{ERA5}}(\lambda)$ and $\bar{I}^{\mathrm{ERA5}}(\lambda)$ within the longitude range of the Eastern Hemisphere from May 9 to 23, 2023 (Fig. 10). The blocking occurrences and the NCCV center positions determined by $I^{\mathrm{ERA5}}(\lambda)$ and $\bar{I}^{\mathrm{ERA5}}(\lambda)$ (Fig. 10) are similar to those in Fig. 9. By comparing the values of blocking indices between Figs. 9 and 10, the variation range of $I^{\mathrm{sat}}(\lambda)$ is smaller than that of $I^{\mathrm{ERA5}}(\lambda)$, with the minimum value of $I^{\mathrm{sat}}(\lambda)$ being greater than that of $I^{\mathrm{ERA5}}(\lambda)$. This is why we changed the threshold to 40 gpm for $I^{\mathrm{sat}}(\lambda)$ and $\bar{I}^{\mathrm{sat}}(\lambda)$ in Fig. 9, instead of zero in Fig. 10. It is found that the longitudes with blocking occurrences determined by [$I^{\mathrm{sat}}(\lambda)$, $\bar{I}^{\mathrm{sat}}(\lambda)$] and [$I^{\mathrm{ERA5}}(\lambda)$, $\bar{I}^{\mathrm{ERA5}}(\lambda)$] are consistent. The AMSU-A temperature sounding channel retrieved Z_{500}^{sat} has excellent performance in detecting high pressure. It should also have some indicative significance for describing other characteristics of atmospheric circulation patterns in the middle troposphere.

To increase the credibility of the proposed method for using satellite microwave temperature observations of brightness temperature to detect blocking occurrences, we show another

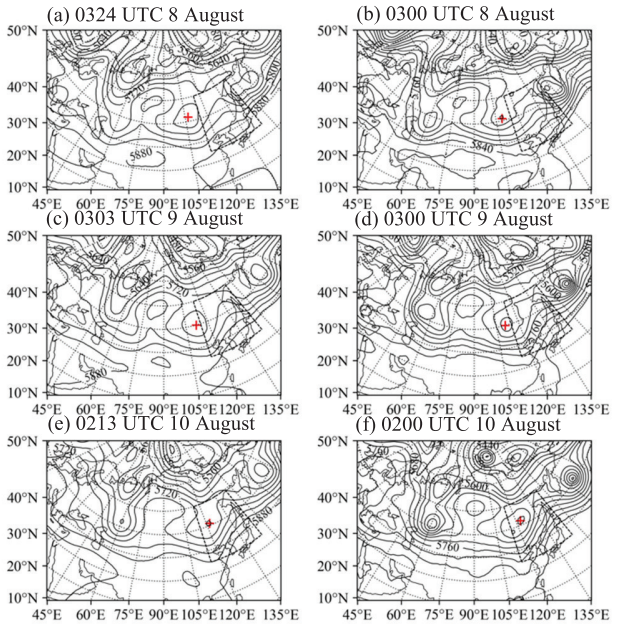


Fig. 11. Spatial distributions of (left panels) Z_{500}^{sat} and (right panels) Z_{500}^{ERA5} (contour interval: 40 gpm) on (a) and (b) August 8, (c) and (d) August 9, and (e) and (f) August 10, 2020. The observing times of MetOp-C AMSU-A brightness temperatures at the domain center and the analysis times are indicated above left and right panels, respectively. The NCCV center position is indicated by a red plus symbol and the NCCV domain by black dashed box.

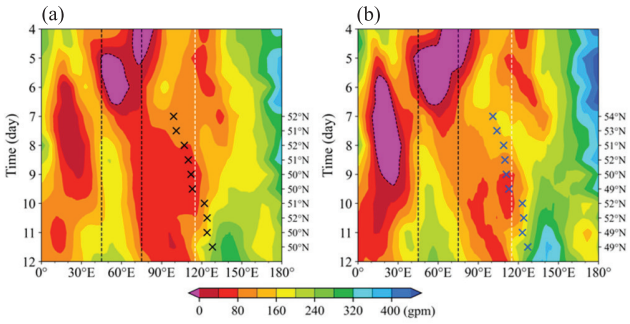


Fig. 12. Temporal variations of (a) (color shading) I^{sat} and (black cross) NCCV center and (b) (color shading) I^{ERA5} and (blue cross) NCCV center during the period from August 4 to 11, 2020 at the observing times of MetOp-C AMSU-A. The longitudes of the Ural and the west edge of the NCCV area are represented by black and white dashed lines, respectively. The black dashed curves in (a) and (b) outline areas within which I^{sat} and I^{ERA5} are less than zero.

blocking/NCCV case in Figs. 11–14. The training data period is from July 21 to August 3, 2020. Fig. 11 presents the spatial distributions of Z_{500}^{sat} and Z_{500}^{ERA5} on August 8–10, 2020. The NCCV was located downstream of a well-developed ridge and to the west and outside the NCCV domain on August 8. It further approached the NCCV area on August 9. By August 10, the NCCV moved into the NCCV domain. Compared with the ERA5 reanalysis, the large-scale flow patterns on 500 hPa are well captured by Z_{500}^{sat}, but the troughs and isolated low systems are not as deep. This is expected as the resolutions (i.e., the FOV sizes) of AMSU-A observations are lower than the ERA5 0.25° × 0.25° resolutions. Figs. 12–14 are similar to Figs. 8–10 except for the second

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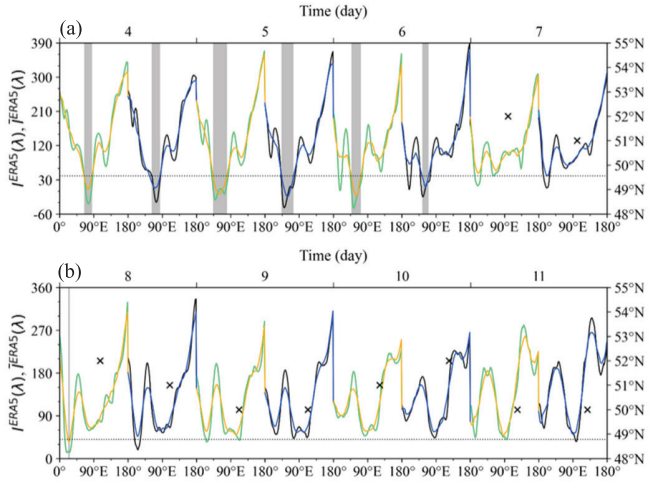


Fig. 13. Same as Fig. 9 except for the second blocking case that occurred during the period from (a) August 4 to 7 and (b) August 8 to 11, 2020.

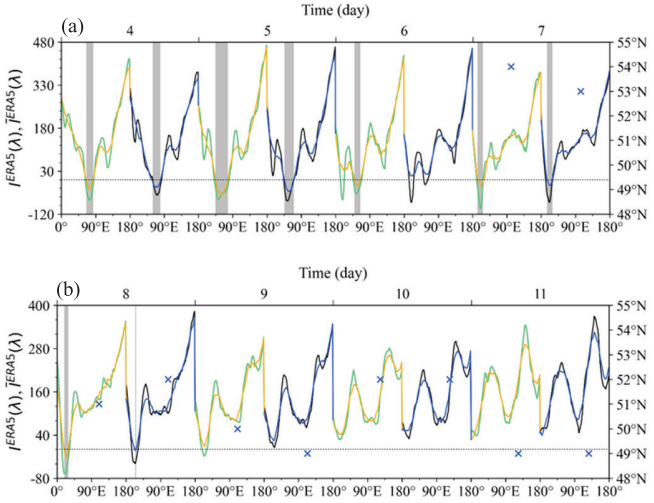


Fig. 14. Same as Fig. 10 except for the second blocking case that occurred during the period from (a) August 4 to 7 and (b) August 8 to 11, 2020.

case. The blocking high pressure system was located over the Ural Mountains region on August 4–6, 2020; and the NCCV formed on August 7 after the active blocking period and moved southeastward in the following five days from August 7 to 11, 2020 (Fig. 12). The geographical locations and times of blocking activity detected by indices I^{sat} and I^{ERA5} (Fig. 13) compare well with those detected by $I^{\text{ERA5}}(\lambda)$ and $\bar{I}^{\text{ERA5}}(\lambda)$ on August 4–6, 2020 (Fig. 14). This confirms that the 40-gpm threshold is appropriate for using the indices $I^{\text{sat}}(\lambda)$ and $\bar{I}^{\text{sat}}(\lambda)$ to objectively detecting the blocking occurrences of the second case.

V. Summary and Conclusion

This research shows that AMSU-A brightness temperature measurements can be directly applied to the study of blocking activities. The MetOp-C AMSU-A temperature sounding channels 4–8 are used to retrieve the 500-hPa geopotential height (Z_{500}^{sat}) for the study of two blocking events that lead to NCCV occurrences. Results show that the spatial distributions of Z_{500}^{sat} compare favorably with the 500-hPa geopotential height from the ERA5 reanalysis (Z_{500}^{ERA5}) in characterizing the circulation patterns of troughs and closed low-pressure systems and ridges. Using two traditional block indices and a modified threshold, AMSU-A-based blocking detection is as successful as the ERA5 reanalysis. This study also demonstrates an innovative application of the three-orbit constellation of FY-3E, MetOp-C, and FY-3D for tracking the NCCV movement paths at the intervals of 3.5, 4, or 4.5 h.

There are more than 26 years of microwave temperature sounding observations since the launch of NOAA-15 on May 13, 1998. In the future, we plan to apply deep learning techniques to the retrievals of geopotential height fields at 500 hPa and other altitude layers, as well as NCCV systems based on observational data of multiple microwave temperature sounders onboard different POES. By then, we may conduct comprehensive and in-depth research on the blocking events, NCCVs, and relationships between the two.

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